Tello Leg: The Study of Design Principles and Metrics for Dynamic Humanoid Robots

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***Abstract*—To be useful tools in real scenarios, humanoid robots must realize tasks dynamically. This means that they must be capable of applying substantial forces and also mitigating impacts that may occur during the motion. Towards creating capable humanoids, this letter presents the leg of the robot Tello and demonstrates how it embodies two new fundamental design concepts for dynamic legged robots. The limbs follow the principles of: (i) Cooperative Actuation (CA), by combining motors in differential configurations to increase the force capability of the limb. We demonstrate that the CA configuration requires half the motor torque to perform a jump in comparison to conventional serial design configurations. And (ii) proximal actuation, by placing heavy motors near the body to reduce the inertia of the limb. To quantify the effect of motor placement on the robot's dynamics, we introduce a novel metric entitled Centroidal Inertia Isotropy (CII). We show that the design of state-of-the-art dynamic legged robots empirically reduces the range of CII to improve agility and facilitate model-based control. We hope this metric will enable a quantifiable way to design these machines in the future.**

***Index Terms*—Actuation and joint mechanisms, humanoid and bipedal locomotion, legged robot.**

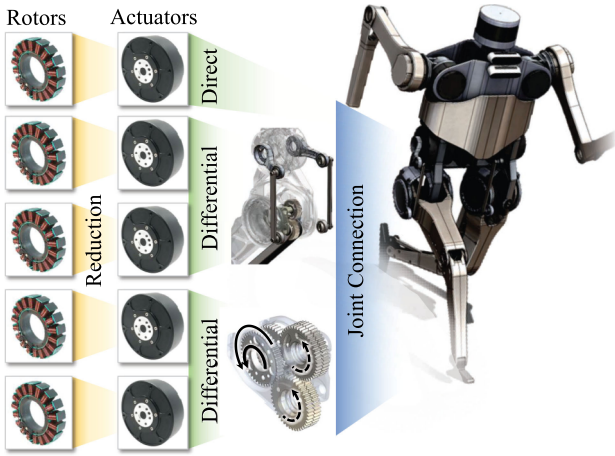


Fig. 1. Physical layers of actuation system in the leg of the humanoid Tello. Motors are arranged in differential configurations to increase the limb's force capability without substantially increasing reflected inertia.

I. Introduction

CAPABLE humanoid robots could be directly inserted in environments designed for humans without major structural modifications. These machines could help firefighters in emergency response or help workers move heavy boxes within warehouses. However, to be capable of realizing physical tasks with human-level performance, humanoid robots must be *dynamic*. Dynamic whole-body movements enable robots to amplify the forces that they can apply to the environment. For instance, the fast movement performed by weightlifters allows them to raise a heavy barbell over their heads, a feat that would be impossible to realize semi-statically. Designing robots that realize tasks dynamically enables minimizing the robot size and mass, making them safer to interact with humans and more cost effective.

Realizing dynamic tasks require operation at the boundaries of the robot's performance. But extending these boundaries requires the formalization of *principles* that guide the design of candidate hardware and *metrics* that enable the quantification of their performance. Towards this goal, this work introduces a novel design principle entitled *Cooperative Actuation (CA)* and a novel performance metric we name *Centroidal Inertia Isotropy (CII)* that are tailored to dynamic legged robots. These concepts were employed in the design of the leg of the dynamic humanoid Tello. Here we show experimental results that demonstrate their use.

On one hand, *design principles* are rules that guide the process of conceiving the hardware of the robot. For instance, Seok [1] demonstrated that employing electric actuators with large gap radius and low gearing improve running efficiency of legged robots. Kim [2] built backdrivable cable-driven manipulators that are safe for physical human robot interaction by reducing contact inertia. Sim [3] and Kikuuwe [4] showed that employing energy efficient mechanical transmissions reduces the reflected inertia of limbs and improve backdrivability. For cable-driven robots, Atsushi [5] reduced torque requirement per actuators with coupling mechanism by combining multiple actuators' torque on joints. Similarly, the CA is a design principle for legged robots for decreasing the torque requirement per motor without inflating the reflected inertia of the limb. This is achieved by combining multiple motors to simultaneously drive multiple joints of the robot's limb that are often not active at the same time. This is in contrast with the common approach of increasing the gearing ratio of transmissions in order to increase the output

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torque of the actuators, which quadratically increases its reflected inertia. The Tello leg employs a pair of actuators to drive the knee and ankle, and a second pair to drive 2°-of-freedom (DoF) hip joint.

On the other hand, *performance metrics* quantify the robot's ability to meet a task requirement. These metrics also work as a common language to compare the performance of different machines. For instance, Wensing [6] introduced the Impact Mitigation Factor (IMF) that quantifies the energetic losses due to inertial loads when the robot's limb impacts the ground. Kojima [7] proposed an evaluation index that measures bending moment and mass of structural frames per actuator torque and significantly reduced frame mass of a humanoid (JAXON3-P) without losing structural strength. This letter introduces the CII as a metric for estimating how closely the robot's full dynamics can be simplified by its centroidal dynamics model [8]. Reduced-order models are widely used with further simplification for planning and control of highly dynamic behaviors for legged robots due to its simplicity for fast computation and competence to capture the core dynamics of the full robot [9]–[16]. For instance, in [17] the Mini Cheetah performs agile running using a combination of a model predictive control (MPC) based on a simplified model with cascaded whole-body control (WBC). Because the robot's limbs are extremely lightweight, the model can accurately represent the full robot dynamics, allowing a WBC controller to easily map the MPC trajectory to the full robot at high control rates around 1 kHz. Although reduced-order models have also been extensively used for planning and control of dynamic humanoid robots [15], [16], it is not as accurate to represent the behavior of these robots due to substantial mass of their limbs. But how lightweight should the limbs of humanoid robots be so the centroidal model can accurately describe the robot's full dynamics? To answer this question, we introduce the CII as a condensed metric to quantify the error between the centroidal model and the full dynamics of the robot.

The fundamental contributions of this letter are the design principle of CA and the performance metric CII tailored to dynamic humanoid robots. We demonstrate these concepts with the design of the leg of the humanoid Tello. In Section II, we discuss a design principle for improving force density and reducing reflected inertia of legged robots by analyzing different configuration of actuators to joint connection. In Section III, a performance metric, CII, that quantifies the proximodistal distribution of mass in legged robot is introduced. We demonstrate that the CII can be evaluated for different types of legged robots and eventually classify robots. In Section IV, a 5-DoF humanoid leg, Tello, is presented as an embodiment of CA to improve joint torque. The 5 actuators of this leg are placed adjacent to the torso such that distal mass is dramatically reduced. In Section V, an experiment of Tello jumping is analyzed to demonstrate the high joint torque enabled by CA.

II. Cooperative Actuation

To realize dynamic tasks, the robot's actuation system needs to deliver large forces when compared to the machine's total mass. In other words, the robot's limbs need to have *high force density*.

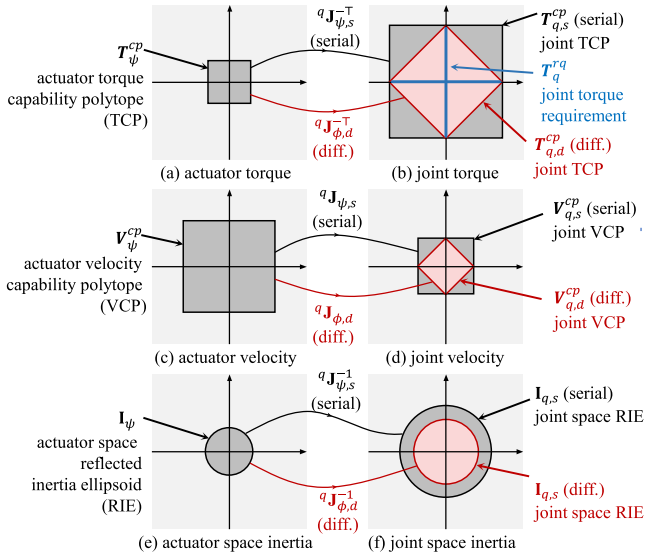


Fig. 2. Torque capability polytopes (top row) and velocity capability polytopes (center row) of actuators are projected onto joint-space. The reflected inertia of the rotors of actuators are depicted as ellipse in actuator-space and joint-space (bottom row).

A straightforward approach to increase the torque limits τ_ψ of actuators is to increase the gearing ratio N of the transmission ($\tau_\psi \propto N$). However, increasing the ratio N increases friction in the joints and quadratically increases the reflected inertia of the rotors ($I_{\text{ref}} \propto N^2$), which degrades the limb's backdrivability and its ability to regulate contact and mitigate impacts [3], [6]. Hence, gearing ratio cannot be increased indefinitely because backdrivability degrades faster than the increase in torque (force) density. As to this motor-gearbox-level design problem, the proprioceptive actuation paradigm [6] suggests that, if a mass budget for an electric actuator is fixed, using actuators with larger gap radius allows less gearing and hence monotonically increases torque density and impact mitigation capability. In contrast, within transmission-level (e.g., actuator-joint or actuator-end-effector), *an alternative solution to increase the force density of the limb with fixed motor mass and size is to combine multiple motors in parallel to simultaneously drive multiple joints*. Parallel configuration allows the joint torque to increase proportionally to the number of motors while just linearly increasing the reflected inertia of the joints. This approach is especially beneficial when these joints are not often recruited simultaneously. Here we study the design trade-offs of combining multiple actuators in differential configuration, which is a particular case of parallel configuration.

To illustrate the benefit of combining multiple actuators, we study the effect of the *topology Jacobian* which represents the connectivity between actuators and joints. For serial manipulators, because each joint is driven by a single actuator, this Jacobian ${}^{q}\mathbf{J}_{\psi,d}$ is simply given by the diagonal matrix whose entries are reciprocals of gear ratio. However, when combining two actuators in a *differential* configuration to drive a pair of joint, each joint is always driven by both actuators:

$$q_1 = \frac{1}{2N_d}(\psi_1 + \psi_2), \quad q_2 = \frac{1}{2N_d}(\psi_1 - \psi_2), \tag{1}$$

where N_d is the gear ratio for differential configuration, q_1, q_2 are joint displacements and ψ_1, ψ_2 are actuator displacements. Hence, the mapping from actuator velocity $\dot{\boldsymbol{\psi}}$ to joint velocity $\dot{\boldsymbol{q}}$ called topology Jacobian ${}^{q}\mathbf{J}_{\phi} : \dot{\boldsymbol{\psi}} \mapsto \dot{\boldsymbol{q}}$ of differential and serial configurations are,

$$ {}^{q}\mathbf{J}_{\psi,d} = \begin{bmatrix} \frac{1}{2N_d} & \frac{1}{2N_d} \\ \frac{1}{2N_d} & -\frac{1}{2N_d} \end{bmatrix}, \quad {}^{q}\mathbf{J}_{\psi,s} = \begin{bmatrix} \frac{1}{N_s} & 0 \\ 0 & \frac{1}{N_s} \end{bmatrix}, \tag{2} $$

where N_s is the gear ratio of actuators in serial configuration. Unlike velocities' transformation, torques of actuators $\boldsymbol{\tau}_{\psi}$ are converted to joint torques $\boldsymbol{\tau}_q$ by inverse of dual maps ${}^{q}\mathbf{J}_{\psi}^{-\top} : \boldsymbol{\tau}_{\psi} \mapsto \boldsymbol{\tau}_q$, which gives torque conversion Jacobians for differential and serial configurations as follows,

$$ {}^{q}\mathbf{J}_{\psi,d}^{-\top} = \begin{bmatrix} N_d & N_d \\ N_d & -N_d \end{bmatrix}, \quad {}^{q}\mathbf{J}_{\psi,s}^{-\top} = \begin{bmatrix} N_s & 0 \\ 0 & N_s \end{bmatrix}. \tag{3} $$

The effect of Jacobians can be visually understood by introducing capability polytopes. The torque and velocity limits of individual actuators are collected as vectors of torque and velocity to form a torque or a velocity capability polytope (TCP, VCP). In serial configuration, the actuator TCP $\boldsymbol{T}_{\psi}^{cp}$ is scaled up to joint TCP $\boldsymbol{T}_{q,s}^{cp}$ by the factor of gear ratio N_s which is described as grey boxes in Fig. 2(a) and (b). In contrast, with differential configuration, the actuator TCP is not only scaled by N_d, but also *rotated* 45 degrees to yield joint TCP $\boldsymbol{T}_{q,d}^{cp}$ which is described as a red box in Fig. 2(b).

The differential configuration can reduce reflected inertia if torques are expected to be applied on a single joint at a time. For example, the knee and ankle torque in bipedal gait do not peak simultaneously, but the concentration of joint torque slide from knee to ankle [18]–[22]. Such joint torque usage is roughly modeled as $\boldsymbol{T}_{q}^{rq}$ as blue lines in Fig. 2(b). For differential configuration to cover the $\boldsymbol{T}_{q}^{rq}$, the polytope is required to inflate *less* than serial configuration because it rotates to the $\boldsymbol{T}_{q}^{rq}$ more efficienctly. Hence, the gear ratio required differential configuration is the half of gear ratio for serial configuration to achieve the same required joint torque,

$$ N_d = \tfrac{1}{2} N_s. \tag{4} $$

Consequently, the joint-space reflected inertia is smaller in case of differential configuration as in Fig. 2(f). The reflected inertia on the join space ${}^{q}\mathbf{I}_{\psi}$ is given as,

$$ {}^{q}\mathbf{I}_{\psi} = {}^{q}\mathbf{J}_{\psi}^{-\top\psi}\mathbf{I}_{\psi}\,{}^{q}\mathbf{J}_{\psi}^{-1}, \tag{5} $$

where ${}^{\psi}\mathbf{I}_{\psi}$ is the rotor inertia matrix whose entries are individual inertia of rotors. The joint-space reflected inertia in differential ${}^{q}\mathbf{I}_{\phi,d}$ and serial configuration ${}^{q}\mathbf{I}_{\phi,s}$ are,

$$ {}^{q}\mathbf{I}_{\phi,d} = \begin{bmatrix} 2N_0^2 & 0 \\ 0 & 2N_0^2 \end{bmatrix}, \quad {}^{q}\mathbf{I}_{\phi,s} = \begin{bmatrix} 4N_0^2 & 0 \\ 0 & 4N_0^2 \end{bmatrix}, \tag{6} $$

where $N_0 := N_d^* = \frac{1}{2} N_s^*$ is used to conveniently compare both inertia tensors. This result indicates that the principal components of reflected inertia of serial configuration are twice larger than those in differential configuration.

The differential configuration also affects velocity conversion between joints and actuators such that there are cases where actuators need to run faster than the joints. In other words, when velocity capability of actuators are given identical to both configurations, differential configuration renders smaller velocity capability on the joint-space as described in the conversion of VCP, $\boldsymbol{V}_{\psi}^{cp}$ to $\boldsymbol{V}_{q,d}^{cp}$ as in Fig. 2(c) and (d).

A. Design Principle: Cooperative Actuation

To develop an actuation system such that generates sufficiently large force and minimizes reflected inertia, two design considerations are joined together as a design principle called *Cooperative Actuation*. First, it is assumed that joint torques are maximally recruited only on a single joint at a time. In other words, the torques of multiple joints should not peak simultaneously. Second, a combination of actuators can work in parallel to drive multiple joints. In this way, load on a joint can be shared by multiple actuators. For example, differential configuration combines actuators' torques so that twice of maximum actuator torque can be applied on a joint. The differential actuation is advantageous under previous assumption because it yields smaller reflected inertia. For serial configuration to yield the maximum joint torque attained by differential configuration, the gear ratio for serial configuration has to be the double of gear ratio for differential configuration as in (4). Such gear ratios yield joint-space reflected inertia of serial configuration twice larger than the joint-space reflected inertia of differential configuration.

III. Centroidal Inertia Isotropy

The controllers of dynamic legged robots such as Cassie, MIT Mini Cheetah, or Atlas employ planners based on reduced-order models with further simplification such as variations of the linear inverted pendulum (LIP) models [9]–[11] spring loaded inverted pendulum (SLIP) model [12], single rigid body model (SRBM) [13], [14] or centroidal dynamics [15], [16]. However, the actual dynamics of legged robots differs from these simple models, because the mass of the limbs contributes to inertia of the whole system. To reduce this difference, aforementioned robots are built with lightweight limbs to reduce the gap between robots' actual dynamics and their simplified models. *This design principle has been empirically used in the design of dynamic legged robots to facilitate control and improve agility, but has never been quantified.*

Although the proximodistal distribution of robot's mass ultimately limits the frequency at the planning runs [17], [23]–[27], the relation between mass distribution and performance of controllers is still not well understood. Hence, it is difficult to draw a line as to how much proximal actuation is needed to attain control objective. In case of humanoid robots, it is more challenging to achieve proximal actuation. First, placing more than 10 electrical motors around hip requires extensive space optimization and more stages of transmission that delivers actuator outputs to distal joints. Moreover, the inertia of the structural frame is larger than quadrupeds due to longer limbs.

This section introduces a design metric, called *Centroidal Inertia Isotropy*, to quantify the state of proximodistal mass

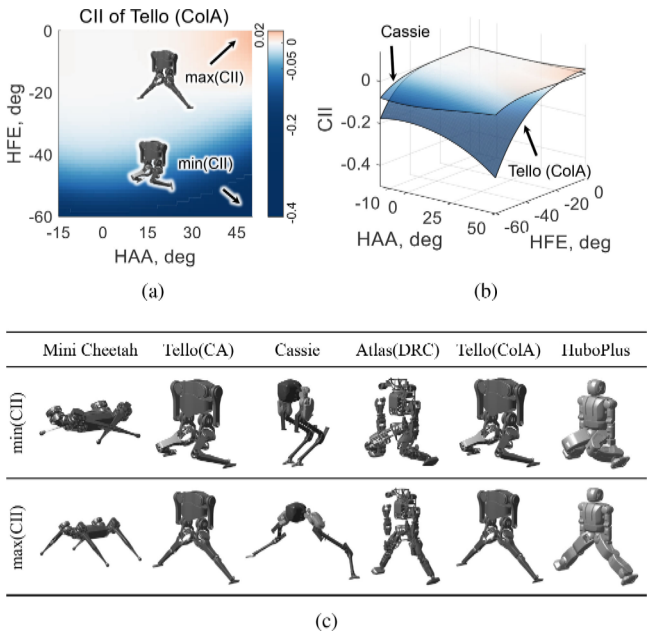


Fig. 3. (a) The CII Tello(ColA) at test configurations; HAA and HFE angles were varied. (b) The CII of Tello(ColA) and Cassie were compared. In case of Tello(ColA), swing mass of actuators mounted on joints contribute to larger CII variation. (c) A class of robots employing proximal actuation and collocated actuation are compared.

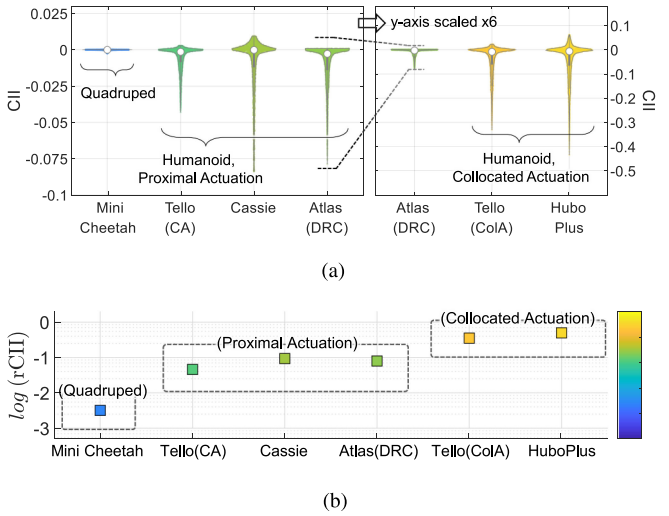


Fig. 4. (a) Various robots at configurations where CII is maximum or minimum. (b) The logarithm plot of rCII reveals the difference in robot types (e.g. humanoid vs quadruped) and groups (e.g. proximally actuated humanoids vs humanoids whose actuators are near the joint).

distribution as a caliber of proximal actuation. The foundational logic behind this metric is that the mass distribution is not directly measured, but the change in inertia of the whole system due to pose (joint configuration) change is quantified, which gives the name inertia isotropy. For example, if a robot has massless limbs, the inertia of the whole system will remain the same regardless of joint configuration. In other words, the inertia is perfectly isotropic on the joint-space. In contrast, the more the mass is distributed toward the distal ends of limbs, the more pronounced is the difference in the inertia of the whole system at different body configurations. The concept of distance between two different inertia matrices is formulated with centroidal dynamics. Consider a robot at nominal configuration $\boldsymbol{q}_0$ with centroidal angular momentum $\boldsymbol{h}_G \in \mathbb{R}^3$ in task-space,

$$\boldsymbol{h}_G = \mathbf{I}_G(\boldsymbol{q}_0)\mathbf{v}_0, \tag{7}$$

where $\mathbf{I}_G \in \mathbb{R}^{3\times3}$ is a submatrix of the original 6×6 centroidal composite rigid body inertia (CCRBI) matrix in [8] which accounts for the rotational inertia (RCCRBI), and $\mathbf{v}_0 \in \mathbb{R}^3$ is a generalized angular velocity vector in configuration $\boldsymbol{q}_0$. Note that subscript G associates its parent vector to a coordinate frame at the point of center of mass (CoM) in task-space.

Next, consider the robot changing its joint configuration to $\boldsymbol{q}$ without external wrench. The RCCRBI in configuration $\boldsymbol{q}$ is derived using the principle of momentum conservation,

$$\boldsymbol{h}_G = \mathbf{I}_G(\boldsymbol{q}_0)\mathbf{v}_0 = \mathbf{I}_G(\boldsymbol{q})\mathbf{v}. \tag{8}$$

If limbs were massless and it is only the torso contributing to RCCRBI, both RCCRBI and velocities would remain unchanged after any change in configuration. However, if limbs are *not* massless, the change from $\boldsymbol{q}_0$ to $\boldsymbol{q}$ leads to difference in generalized velocities, $\mathbf{v}$ and $\mathbf{v}_0$,

$$\mathbf{v} - \mathbf{v}_0 = (\mathbf{I}_G^{-1}(\boldsymbol{q})\mathbf{I}_G(\boldsymbol{q}_0) - \mathbf{1}_3)\mathbf{v}_0, \tag{9}$$

where $\mathbf{1}_3 \in \mathbb{R}^{3\times3}$ is the identity matrix. After normalizing the difference in the velocities by the generalized velocity at nominal configuration $\mathbf{v}_0$ and taking the determinant of the coefficient matrix, the CII is obtained,

$$\mathrm{CII}\,(\boldsymbol{q}, \boldsymbol{q}_0) := \det\left(\mathbf{I}_G^{-1}(\boldsymbol{q})\mathbf{I}_G(\boldsymbol{q}_0) - \mathbf{1}_3\right). \tag{10}$$

One notable characteristic of CII is that CII is a local index; CII is evaluated per a joint configuration. Hence, the isotropy in RCCRBI is measured by the distribution of CII evaluated over a set of configurations. In this study, CII of various robots such as quadruped and humanoids that are built with different paradigms are evaluated.[1] To highlight the difference between proximally actuated robots and robots whose actuators are collocated on joint, two humanoid models were included in the study. First, the Tello(CA) uses differential mechanisms and has all actuators placed around hip joint which will be discussed in Section IV. Second, Tello(ColA)(Collocated Actuation) employs serial mechanism and actuators are collocated on corresponding joints. The nominal configuration is defined as straight-leg and straight-up pose. The set of test configurations $\boldsymbol{Q} \in \mathbb{R}^2$ was populated by changing two variables: hip abduction/adduction (HAA) angle and hip flexion/extension (HFE) angle. The knee angle is chosen such that the ankle stays on the frontal plane. As a preliminary result, Fig. 3 displays the distribution of 50×50 grid of HAA $\in [-15, 50]$ deg and HFE $\in [-60, 0]$ deg. This study limits i) hip adduction to prevent self-collision of two legs and ii) HFE to be nonpositive to allow either backward-knee or forward-knee configuration only.

[1]The details of robots and mechanics of CII are available in https://t.ly/Of7P.

A few notable observations are, firstly, the range of CII (rCII) shows order-difference across different groups of robots. The rCII is defined as the difference between maximum and minimum CII for a given configuration set $\boldsymbol{Q}$ as follows,

$$\mathrm{rCII}(\boldsymbol{Q}) := \mathrm{CII}(\bar{\boldsymbol{q}}, \boldsymbol{q}_0) - \mathrm{CII}(\underline{\boldsymbol{q}}, \boldsymbol{q}_0), \tag{11}$$

where $\boldsymbol{q}_0, \bar{\boldsymbol{q}}, \underline{\boldsymbol{q}} \in \boldsymbol{Q}$ are nominal configuration, and configurations where CII is maximum and minimum. The group of robots designed from the principle of proximal actuation (e.g., MIT Mini Cheetah, Tello(CA), Cassie, Atlas(DRC)) is easily differentiated from a group of robots whose actuators are placed near corresponding joints (e.g., Tello(ColA) and HuboPlus). Within the proximal actuation group as in Fig. 4(a), the humanoids show significantly larger rCII compared to the quadruped. The difference in rCII and the idea of differentiating a class of robot is more obviously demonstrated by taking the logarithm of rCII as in Fig. 4(b). Second, visually comparing the configurations where CII is maximum or minimum (Fig 3(c)), a clear trend in posture were observed across robots. In minimum CII configuration, the robots crouch to fit inside a ball, while they tend to extend their limbs as much as possible in maximum CII configuration.

IV. Implementation of Tello Leg

A. Cooperative Actuation for Hip, Knee and Ankle Joints

A 5-DoF humanoid leg, called Tello, is designed to maximize joint torque with minimal reflected inertia. The actuation system of the Tello leg was built upon the idea of cooperative actuation discussed in Section II. The core concept of CA is that, if joint torques are not recruited simultaneously, it is beneficial to share loading on a joint with multiple actuators such that torque requirement on the actuators are reduced. In this way, the reflected inertia is rendered smaller, in comparison to employing serial configuration with higher gearing ratio. The 2-DoF cooperative actuation is realized in the knee and ankle joint pair. Specifically speaking of bipedal locomotion, multiple studies state that joint torque of the knee and ankle do not peak simultaneously during nominal walking in case of healthy humans [18], humans with lower limb prosthesis [20], simulated gaits of bipedal robots [21]. This pattern provides basis to the idea of dynamically shifting the concentration of actuators' torque onto a single joint when needed.

Another case of 2-DoF cooperative actuation is employed in the control of hip abduction and adduction (HAA) and hip flexion and extension (HFE) to promote space optimization to place actuators closer to the CoM and to double the maximum joint torque at the cost of doubled reflected inertia.

B. Transmission for 5-DoF Proximal Actuation

Towards minimizing rCII, all 5 motors of Tello are placed around or above its hip joint, as depicted in Fig. 5(b) and their motion and torques are transmitted through a series of mechanisms. For a humanoids leg, Tello leg exhibits comparable CII to other recent humanoids as in Fig. 3(b).

The first CA is established between a pair of joints and a pair of motors; knee and ankle joint represented as q_4, q_5 in Fig. 5(a)

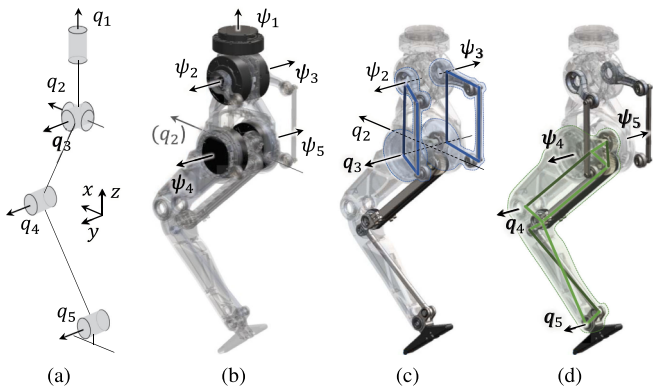


Fig. 5. (a) Stick figure of joint connection (b) Placement of 5 motors (c) Hip differential mechanism (d) Knee and ankle differential mechanism.

and actuator 4 & 5 whose axes are labed as ψ_4, ψ_5 in Fig. 5(b). The movement of the two motors are distributed to two routes by a spur gear differential. One route connects to knee actuation via hamstring-like parallel mechanism in Fig. 5(c) and the other route connects to ankle actuation via a 4-bar linkage located on the shank. The kinematics of knee and ankle differential pair is given as,

$$\begin{aligned} q_4 &= f_4(\psi_4, \psi_5) = \tfrac{1}{2}\psi_4 + \gamma\psi_5 + q_4^o, \\ q_5 &= f_5(\psi_4, \psi_5) = \tfrac{1}{2}\psi_4 - \gamma\psi_5 + q_5^o, \end{aligned} \tag{12}$$

where γ is a geometric parameter of the spur gear differential whose value is $1/2$ in this particular implementation and $q_{(.)}^o$ are joint offsets.

The second CA is employed between another pair of joints and actuators; HAA q_2 and HFE q_3 as the joint pair and actuator 2 ψ_2 and actuator 3 ψ_3 as the actuator pair. Note that actuators are placed remotely from the joints of actuation as in Fig. 5(b). The non-planer 4-bar linkages used in the second CA results in non-linear kinematics, which is numerically approximated with fifth-order polynomials f_2 and f_3,

$$q_2 = f_2(\psi_2, \psi_3), \qquad q_3 = f_3(\psi_2, \psi_3).$$

Finally, the hip rotation q_1 is directly driven by actuator 1 ψ_1, which gives, $q_1 = \psi_1$.

The full topology Jacobian ${}^q\mathbf{J}_\psi \in \mathbb{R}^{5\times5}$ from motor velocity $\boldsymbol{v}_\psi \in \mathbb{R}^5$ to joint velocity $\boldsymbol{v}_\psi \in \mathbb{R}^5$ is obtained by taking the derivative of above kinematics, $\boldsymbol{q} = \boldsymbol{f}(\boldsymbol{\psi})$,

$${}^q\mathbf{J}_\psi = \frac{d\boldsymbol{f}}{d\boldsymbol{\psi}}, \tag{13}$$

$${}^q\mathbf{J}_\psi\big|_{\boldsymbol{q}=0} = \mathrm{diag}\left(1, \begin{bmatrix} \beta & \beta \\ 1/2 & 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 & \gamma \\ 1/2 & -\gamma \end{bmatrix}\right), \tag{14}$$

where ${}^q\mathbf{J}_\psi|_{\boldsymbol{q}=\mathbf{0}}$ is an example of linearized actuation topology around home position ($\boldsymbol{q} = \mathbf{0}$) and $\beta = 0.465$ is a geometric parameter of the non-planar 4-bar linkage.

V. Experiments

This section shows experimental results of a jump achieved with Tello leg. The goal of this experiment is to validate i) the

CA embodied for the actuation of knee and ankle joint and ii) transmission design for minimizing CII of 5-DoF leg. The Tello leg affixed at the end of a passive 2-DoF boom which allows rotation in elevation and azimuth. The leg was placed on a force place to measure the ground reaction force (GRF) at 1 kHz. In this particular experiment, the first joint (hip yaw) was locked and the foot was removed so that the ankle joint makes contact with the force plate.

A. Hardware Setup

A real-time controller (National Instruments, sbRIO-9627) reads displacements and estimate the velocities of 5 actuators and commands reference torques to BLDC motor drivers (Advanced Motion Control, FM060-CM-25) via analog channel. This control loop runs at 1 kHz. Tello is equipped with quasi-direct drive motors (Steadywin, V3) which contain a single-stage 6:1 planetary gearbox. This actuator has similar properties to those of the Mini Cheetah [28]. The mass of the entire setup including the robot and the boom measures $m_t = 6.9$ kg.

B. Control for Jumping

A state machine administers jumping and landing in following order of state transitions. In the beginning, the Tello maintains *Ground* state where impedance controller keeps the knee bent, so that it can accumulate enough momentum during thrust to make a jump. When a jump is commanded, the state machine is in *Thrust* state where pure feedforward force control is employed. The feedforward force $\boldsymbol{\tau}_x$ has z-directional component only which is prescribed by a fifth-order Bézier curve. Next, when vertical GRF measured from the force plate reaches lower threshold (8 N), the state changes to *Aerial* state where joint positional and differential (PD) control is engaged to retract the foot to prevent falling into singularity (straight leg) and prepare soft landing. Next, upon detecting increase in GRF due to contact, the state turns back into *Ground*. To prevent bouncing between states due to unstable contact, the *Ground* state is locked for 2 s after the first entry from the *Aerial* state to the *Ground* state.

The control law for actuator torques $\boldsymbol{\tau}_\psi \in \mathbb{R}^5$ per state is

$$\boldsymbol{\tau}_\psi = \begin{cases} {}^x\mathbf{J}_\psi^\top \left(m_t \boldsymbol{g} + \mathbf{K}_x^p \delta\boldsymbol{x} + \mathbf{K}_x^d \delta\dot{\boldsymbol{x}}\right) & \text{(Ground)}, \\ {}^x\mathbf{J}_\psi^\top \boldsymbol{\tau}_x & \text{(Thrust)}, \\ {}^q\mathbf{J}_\psi^\top \left(\mathbf{K}_q^p \delta\boldsymbol{q} + \mathbf{K}_q^d \delta\dot{\boldsymbol{q}}\right) & \text{(Aerial)}, \end{cases} \tag{15}$$

where the dimension of Jacobians are ${}^x\mathbf{J}_\psi^\top, {}^q\mathbf{J}_\psi^\top \in \mathbb{R}^{5\times 3}$, $\boldsymbol{g} \in \mathbb{R}^3$ is the gravitational acceleration, $\mathbf{K}_x^p, \mathbf{K}_x^d \in \mathbb{R}^{3\times 3}$ are task-space PD gain matrices, $\delta\boldsymbol{x}, \delta\dot{\boldsymbol{x}}$ are position and velocity error in task-space, $\mathbf{K}_q^p, \mathbf{K}_q^d \in \mathbb{R}^{5\times 5}$ are joint-space PD gain matrices, $\delta\boldsymbol{q}, \delta\dot{\boldsymbol{q}}$ are position and velocity error in joint-space. Next, the individual entries of actuator torque vector $\boldsymbol{\tau}_\psi$ are saturated to $[10, -10]$ Nm in software for safety ($\approx 67\%$ peak motor torque).

C. Results

The Tello leg successfully jumped 171 mm vertically and landed softly as in Fig. 6. The $Thrust$ state took 220 ms and stayed in $Aerial$ state for 275 ms. Note that the actual torques applied on motors were not recorded due to lack of instrumentation. Instead, this analysis assumes that difference between commanded torque and exerted torque are negligible. From hereon, the result is analyzed in chronological order starting from *Thrust* state with Figs. 7 and 8. Next, the benefit and possible disadvantage of CA is analyzed with Fig. 8(b) and (d).

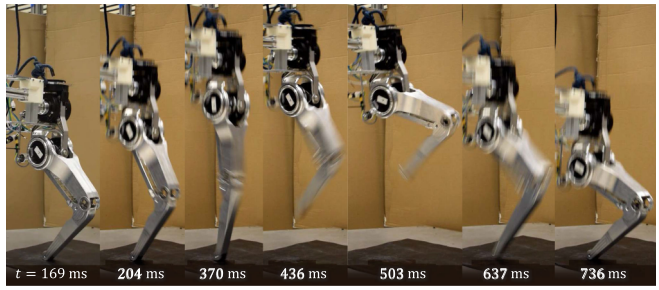


Fig. 6. Jumping in place. With the design philosophy and realization of transmission described in this letter, the robot weighing 6.9 kg jumped successfully with 5 motors attached around its hip.

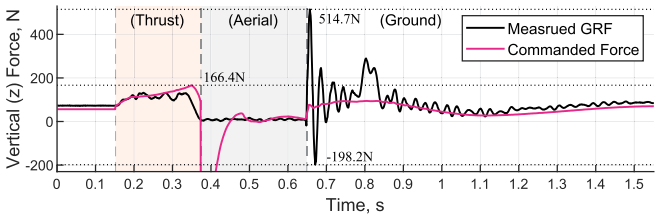


Fig. 7. Commanded vertical force on the foot and vertical GRF measured from a force plate are compared.

In *Thrust* state, the difference between the commanded vertical force and the measured vertical GRF is kept small as in Fig. 7 until 0.3 s mark. This difference grows larger as the leg accelerates towards take-off because i) the controller does not compensate friction or inertial forces needed to accelerate limbs and ii) the angular velocity of actuators increases considerably (e.g., -29.4 rad/s) as shown in Fig. 8(a) which can limit maximum torque applied. Next, the effectiveness of CA, which is to utilize differential configuration to focus multiple actuators' torque on a joint at a time is verified. With actuators whose maximum torque is ± 10 Nm, larger knee torque, -20 Nm, was commanded during the $Thrust$ state (Fig. 8(a)) which is critical for explosive jumping. For a leg to generate thrust force with the CoM vertically aligned with the center of pressure, it is expected that only knee extension torque contributes to the thrust whereas hip and ankle torque do not. The result aligns with this expectation; the hip flexion torque practically remained zero (Fig. 8(c)) and ankle torque was kept relatively small (Fig. 8(a)) with in *Thrust* state.

In *Aerial* state, the joint-space PD controller tries to retreat the leg into a bent-knee pose for smooth landing. Immediately after take-off (0.37 s), the knee is almost straight as in Fig. 6 or the manipulator Jacobian is near singularity. Hence, the projected force on task-space is rendered significantly large (plunge in commanded force at 0.37 s mark in Fig. 7). To retreat the leg

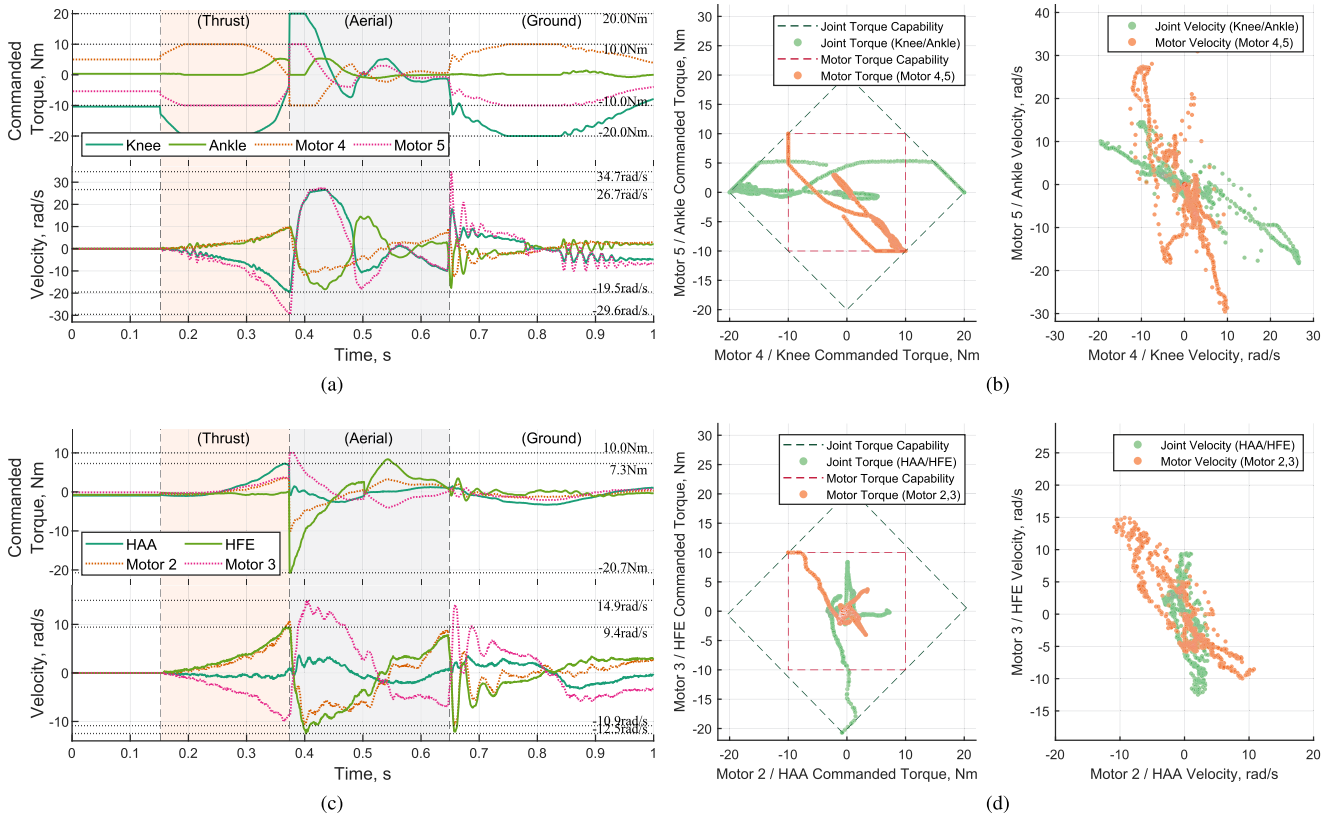


Fig. 8. (a) The trajectories of commanded torques and velocity of a CA pair involving knee, ankle and actuator 4 and 5. (b) The trajectories of torque (left) and velocity (right) of knee & ankle joints and actuators 4 & 5. (c) The trajectories of torques and velocity of a CA pair involving HAA, HFE, actuator 2 and 3. (d) The trajectories of torque (left) and velocity (right) of HAA & HFE joints and actuators 2 & 3.

quickly to prevent landing with straight-knee as much as possible, the controller commands maximum knee flexion torque, $+20$ Nm (Fig. 8(a)), and maximum hip flexion torque, -20.7 Nm (Fig. 8(c)).

Finally, the high frequency ringing in vertical GRF reading in *Ground* state after 0.66 s mark (Fig. 7) is due to oscillation of the force plate incurred by landing impact.

The true benefit of utilizing CA on the knee and ankle joint pair is revealed in Fig. 8(b) from the fact that the span of KFE torque trajectory is wider than that of motor torques. This means that the differential configuration can generate large knee joint torque by combining two actuator torques. The range of commanded ankle torque was kept relatively small during the experiment. However, In contrast to the fact that actuator torques can be added up to create larger joint torque with differential configuration, twice larger velocity than joint velocity may be requested on actuators. For instance, if joint velocities are $(\dot{q}_4, \dot{q}_5) = (v_0, -v_0)$ for arbitrary velocity v_0, then actuator velocities are $(\dot{\psi}_4, \dot{\psi}_5) = (0, -2v_0)$ which can be deduced from the (12). However, during jumping experiment, such occasion was not present and, hence, actuator velocities were not necessarily twice larger than joint velocities. As in Fig. 8(a), the CA employed in knee and ankle joint showed actuator velocities ranging within $[-29.6, 34.7]$ rad/s, while joint velocities were in $[-19.5, 26.7]$ rad/s. In case of CA employed in HFE & HFE joints (Fig. 8(b)), the actuator velocities were within $[-10.9, 14.9]$ rad/s, while joint velocities ranged within $[-12.5, 9.4]$ rad/s.

One miscellaneous comment is that, as the leg stretches to jump, boom generates spherical motion while the leg tries to move vertically. The controller perceives this mismatch as if the ankle is sliding in medio-lateral direction (y-axis). Hence, the task-space force controller gradually recruits larger HAA torque to correct the mismatch (Fig. 8(a)).

D. Discussion

The jump experiment demonstrated that CA employed in the knee and ankle joint pair can recruit two actuators to exert approximately twice larger torque on knee joint than maximum actuator torque. At the same time, the ankle torque was kept small such that majority of the torque outputs of the two actuators are routed onto the knee joint. Towards minimizing rCII, the Tello leg has all five actuators placed near the center of mass. The second case of cooperative actuation which is the differential configuration on the HAA and HFE joint helped optimizing the actuators' placement. The actuators 2 & 3 were placed remotely from the joint such that it allows actuators 4 & 5 which drives knee and ankle joints can be placed on the hip joint. One concern was that differential configuration may require actuators to run faster than joint velocities due to the kinematics of differential mechanism (e.g., twice faster, maximally). Although such occasion was not observed from jump experiment, it may happen in other type of motions such as running. One downside of the transmission design for 5-DoF proximal actuation (Fig. 5) is the complexity of the mechanical build. For instance, the range

of motion of ankle flexion is limited due to the chain of 4-bar linkages that connects actuators placed on the hip to the ankle joint. Still, the CA for ankle and knee operation has shown great benefit. Hence, to counter the drawback of increased mechanical complexity, other space optimization methods and more efficient design choices for parallel transmission will be explored in future studies. Although substantial design effort was invested in the build of Tello, the resultant rCII (Fig. 4(b)) is only slightly better than Cassie, which employs one actuator attached closer to the distal side or right under the tarsus joint. Therefore, authors will investigate in future work if the the transmission design of Tello guarantees ultimately simpler control or superior performance envelope than a robot like Cassie.

VI. Conclusion

The proposed design principle of cooperative actuation which aims to generate large force in bipeds while minimizing reflected inertia so that legged robots can rapidly regulate interaction with environment. We also propose a performance metric CII which quantifies the degree to proximal actuation, or how much mass is concentrated toward the center of mass. The CII was evaluated for different types and class of existing robots and it successfully demonstrated that CII can predict the type of robot and the placement of actuators of a robot. We have built a 5-DoF leg for humanoid robot, Tello, upon the idea of CA. All 5 actuators of the leg are placed near the center of mass to promote more efficient control synthesis and implementation. The Tello leg was able to jump by recruiting multiple motors to amplify joint torques.

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